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SUPERSEDING
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MILITARY SPECIFICATION

TRAINER, MORTAR: M32A1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an air driven mortar trainer for insertion into the bore of a 60MM, 81MM or 4.2 Inch Mortar.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General Specification for
MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging and Packing of

STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions

PUBLICATIONS

U. S. Army Armament Command

Index List 11578263 - Index List for Trainer, Mortar: M32A1.
TM 9-6920-212-14 - Subcaliber Mortar Trainer M32 with 25MM Training Projectile M379.
PDS P8433199 - Packaging Data Sheet Mortar Trainer: M32A1.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified, the first article shall be included in the first article package (see 5.1).

3.2 Materials and construction. The mortar trainer shall conform to this specification, the drawings listed in Index List 11578263 and to MIL-C-13931.

3.2.1 Functioning. The mortar trainer shall function without interference, erratic movement or malfunction when operating inside the mortar tube.

3.2.2 Air leakage. There shall be no leakage of air in any of the valve assemblies, valve fittings or cylinder fittings.

3.3 Modification restrictions. The mounting and operation of the mortar trainer shall not require modification of the operational mortar.

3.4 Accuracy. The mortar trainer shall have an accuracy of placing all projectiles within a six foot radius circle at each range specified when fired in accordance with Tables I, II, or III. These tables require use of the 2000 inch range charge setting cover for the applicable mortar. The center of each circle shall be on the mortar-target line exactly at the specified ranges.

Table I. Firing based on 60MM Mortar

Range (Inches)	Elevation (Mils)	Charge
348	800	0
631	1079	1
910	1156	2
1162	1177	3
1415	1031	3
1718	1053	4
1978	800	4

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Table II. Firing based on 81MM Mortar

Range (Inches)	Elevation (Mils)	Charge
303	1045	0
606	1032	1
909	1011	2
1166	1014	3
1424	997	4
1757	886	5
2000	835	6

Table III. Firing based on 4.2 Inch Mortar

Range (Inches)	Elevation (Mils)	Charge
314	800	4.0
610	885	7.0
915	848	10.0
1174	900	13.0
1456	954	17.0
1750	900	21.0
2000	800	25.5

3.5 Air cylinder. The air cylinders shall be furnished uncharged.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. A first article shall be submitted for inspection in accordance with contract requirements. The first article shall be representative of the production process to be used during quantity production. The first article shall be subjected to the quality conformance inspections specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection provisions and methods. Inspection provisions and methods shall be in accordance with MIL-C-13931 except as otherwise specified herein.

4.5.1.1 Materials. Materials for use in parts shall be inspected in accordance with MIL-C-13931 and the applicable material specifications referenced on the part drawings.

4.5.1.2 Inspection procedures. Parts, subassemblies, and trainers shall be inspected in accordance with this specification and MIL-C-13931 and the applicable Supplementary Quality Assurance Provisions (SQAPs).

4.5.2 Examination.

4.5.2.1 Examination of assembled trainers. Each trainer shall be visually examined for completeness of assembly and correct markings.

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4.5.3 Tests.

4.5.3.1 Air cylinder tests. The following tests on each mortar trainer shall require pressurization of each air cylinder to the specified pressure. At the conclusion of the tests, each air cylinder shall be discharged in accordance with 3.5.

4.5.3.2 Pneumatic component operational test. Connect an auxiliary calibrated gage to the end of the air hose. Open the hand valve on the air cylinder. Verify that the high pressure gage reads 1800 ± 100 psi. With the hand valve open, turn the pressure regulator handle clockwise until it stops. Verify that the auxiliary calibrated gage reads 140 ± 10 psi. Note the position of the needle of the charge setting gage relative to the 150 psi marking of the charge setting gage. Turn the pressure regulator handle counterclockwise until the load on the regulator diaphragm is removed. Verify that the auxiliary pressure gage and the charge setting gage both read zero psi. Close the air cylinder hand valve. The air cylinder valve, the pressure regulator, and their respective gages must operate smoothly throughout their travel, with no binding of the valve stems and no interrupted movement of the gage needles. Failure of the trainer to meet these tests shall require rejection of the unit.

4.5.3.3 Pneumatic component leakage test. With the mortar trainer completely assembled, but not mounted in a mortar, open the hand valve on the air cylinder. Turn the handle on the pressure regulator until the charge setting gage needle reaches approximately half-scale. Paint each of the system fittings with a soap solution, and check for air leaks as evidenced by bubbles forming at the fittings. In a similar manner, check for leakage past the relief valve. No leakage shall be allowed. Leakage in any component shall be cause for rejection of the trainer. Shut off the air cylinder valve and the pressure regulator. At this time as a precautionary measure, actuate the relief valve so that the entire system will be at a no-load condition. Disassemble the barrel from the trainer. Pressurize the system according to the above procedure. Paint the projectile-impact area of the ejection valve with a soap solution and check for air leakage through the valve. No leakage shall be allowed. Leakage shall require rejection of the trainer. Shut off the air cylinder valve and the pressure regulator. At this time as a precautionary measure, actuate the relief valve so that the entire system will be at a no-load condition. Finally, at the completion of these tests, clean all affected parts of the mortar trainer by wiping them dry with a clean, lint-free cloth or other similar material.

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4.5.3.4 Manual operating handle functional test. Mount the trainer to the operational mortar. Pressurize the system so that the charge setting gage needle reaches approximately half-scale. Keep all personnel clear of the muzzle end of the barrel. Strike the manual operating handle a sharp blow with the hand. Verify that the ejection valve is actuated properly as evidenced by the sound of air escaping through the valve and barrel into the open atmosphere. Malfunction of the components involved shall require rejection of the trainer. Shut off the air cylinder valve and the pressure regulator.

4.5.3.5 Relief valve functional test. Following the operating handle functional test, reopen the cylinder valve and the regulator valve until the system pressure reaches full scale (150 psi). Close the cylinder valve and actuate the pressure relief valve. The system pressure should return to zero. Failure to meet this test shall require rejection of the relief valve.

4.5.3.6 Complete mortar trainer operational test. Mount the trainer in the operational mortar and fire two rounds at each of the ranges listed in the applicable table. The 2000-inch-range charge-setting cover for the applicable mortar shall be used. Failure of any rounds to fall within the prescribed circle shall be cause for rejection of the trainer.

4.5.3.6.1 Reduced firing. After 20 consecutive mortar trainers have successfully passed the firing test of 4.5.3.6, reduced firing shall commence. Reduced firing shall be accomplished by firing two rounds each at the 1st (low range), 4th (mid range) and 7th (high range) ranges listed in Tables I, II, or III as applicable. The 2000-inch range charge-setting cover for the applicable mortar shall be used. Failure of any rounds to fall within the prescribed circle shall be cause for rejection of the trainer, forfeiture of reduced firing, and the return to the complete operational tests of 4.5.3.6 until requalification is accomplished. Requalification shall be achieved when 20 consecutive mortar trainers have successfully passed the firing test of 4.5.3.6.

4.6 Operational manual. Refer to TM 9-6920-212-14 for information concerning the operation of the mortar trainer.

4.7 Inspection of preparation for delivery.

4.7.1 Packaging inspection. Unless otherwise specified, inspection to determine compliance with cleaning, preservation, packaging, packing, and marking requirements of the applicable packaging data sheets for the level designated in the contract shall be as specified in MIL-P-14232.

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5. PREPARATION FOR DELIVERY

5.1 First article package. On each contract, a first article package shall be provided in accordance with 3.1. The first article package for the mortar trainer shall be unit packaged for the level of protection specified in the contract and packed level C. Unit packaging and packing shall be in accordance with MIL-P-14232 and Packaging Data Sheet P8433199.

5.2 Unit packaging and packing of the mortar trainer. The mortar trainer shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and PDS P8433199 for the level of protection specified in the contract.

5.3 Unit packaging and packing of repair parts. Repair parts shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and PDS P8433199 for the level of protection specified in the contract.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Requirements for submission of the first article and the first article package, if required.

(c) The applicable packaging data sheet and the level of packaging and packing required.

(d) Serial numbers for the mortar trainers.

(e) Availability of inspection equipment from Government stocks.

(f) Extent of supplier's responsibility for Government-furnished and for supplier-acquired final inspection equipment.

6.2 Projectile. The projectile used in the M32A1 Mortar Trainer is the 25MM, M379 Projectile per MIL-P-46416.

6.3 Contract data requirements. Monthly reports of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

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6.4 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.5 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

Custodian:

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Preparing activity:

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